

Hyperdimensional Probe

Information decoding using **V**ector **S**ymbolic **A**rchitectures



UNIVERSITÀ
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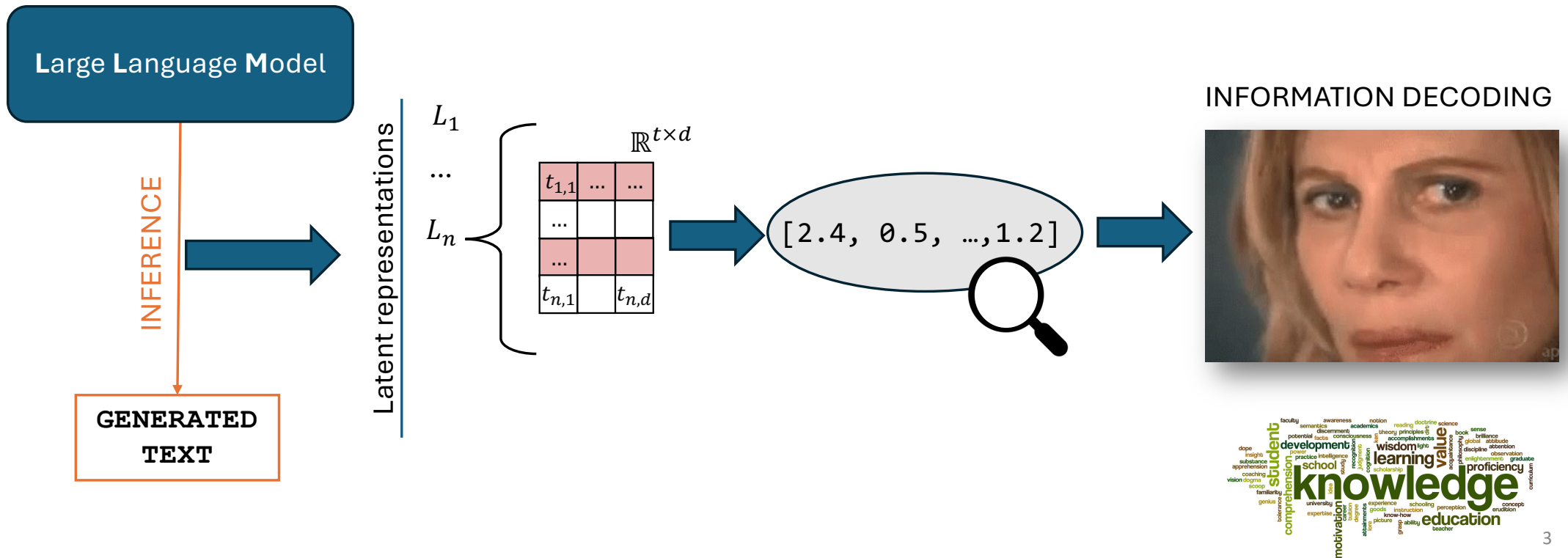


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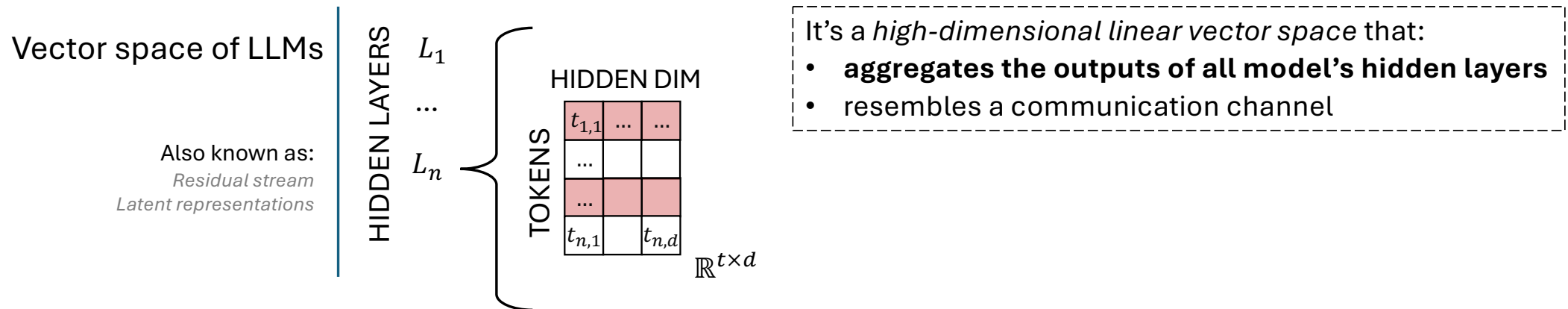
MOTIVATION

There's *limited understanding* of **internal representations** of LLMs



TASK

Information decoding → extract information from the *vector space of transformer models*



TASK

Information decoding → **extract information** from the *vector space* of transformer models

↓
Identification of human-interpretable **features**

- ↓
- a) *Concepts* (e.g., 🐶, 🐼)
 - b) *Linguistic features* (e.g., noun, agent)
 - c) *Abstract properties* (e.g., truthfulness)
 - d) ...

CURRENT APPROACHES (1)

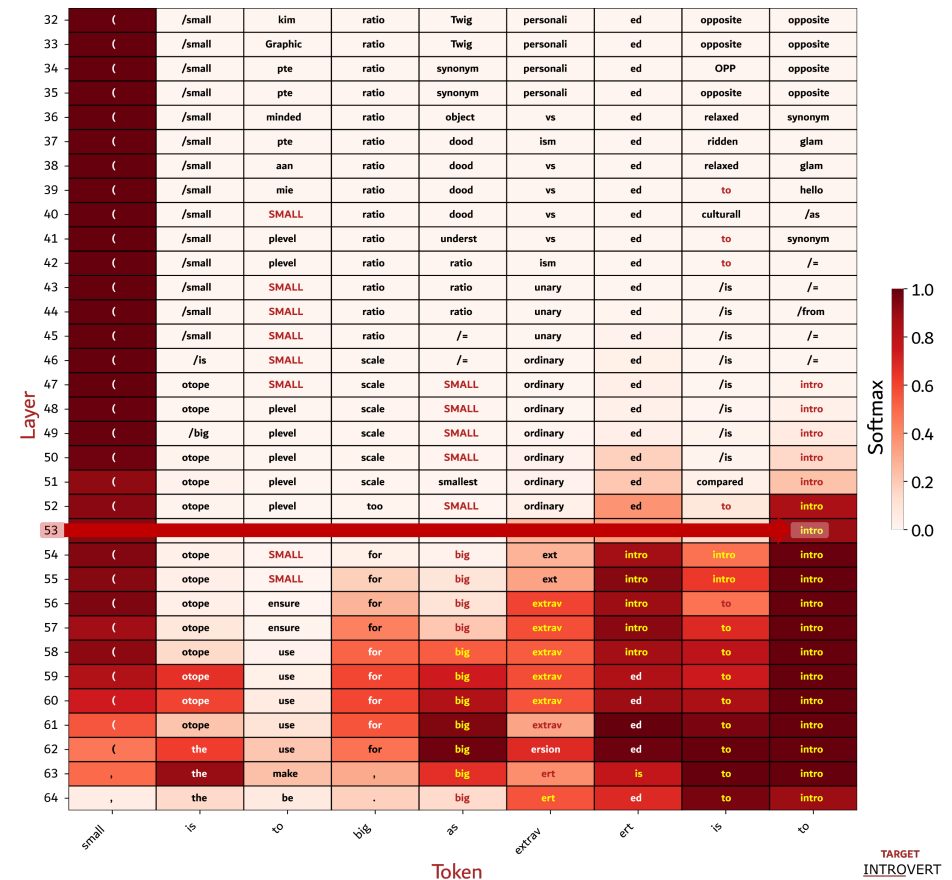
1) Logit Attribution

e.g., Logit Lens

- It projects the model's latent representations onto its **output vocabulary** via the *unembedding layer or affine transformation*.
- It reveals *next-token predictions*, assuming that **all subsequent layers are bypassed**, providing insight into prediction dynamics.

LIMITATIONS

- It focuses only on the **next-token representation**
- Abstraction is **constrained to model's tokens**



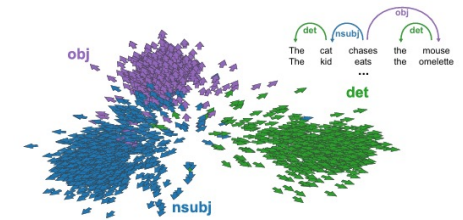
CURRENT APPROACHES (2)

1) Direct Logit Attribution (DLA)

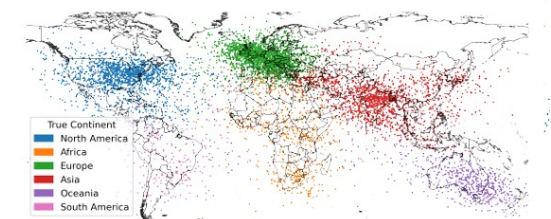
2) Supervised Probes

- a) A supervised model is trained to **map the model's vector space to meaningful features**, such as:
- *syntactical information* (nsubj, obj, det, ..)
 - *space-time coordinates* (latitude, longitude)
 - *abstract properties* (truthfulness)
- b) They aim to **quantity how much information about them is embedded**.

PCA of probed relations



[A polar coordinate system represents syntax in large language models](#)



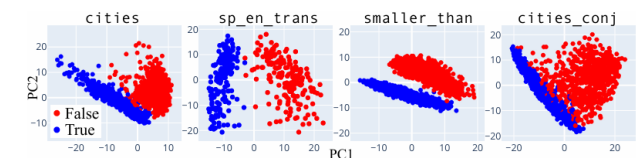
[Language Models Represent Space and Time](#)



CHALLENGE

1. Assumption of *linearity*
2. Difficult to separate **information decoding** from **probe learning**

Was the information encoded or learned by the probe itself?



[The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets](#)

CURRENT APPROACHES (3)

1) Direct Logit Attribution (DLA)

2) Supervised Probes

3) *Sparse Autoencoders* (SAEs)

a) An *autoencoder* **reconstructs the residual stream in an unsupervised fashion**
→ enforcing *sparsity* in its internal representations.

b) Once trained, these representations **serve as a proxy layer to analyse LLM vector space**

c) SAE **activated neurons** are interpreted via two main strategies:

- 1) identify representative ***tokens using logit attribution***
- 2) clustering inputs by ***shared SAE neuron activation***
 - followed by *manual* or *automatic* neuron/feature naming.

CURRENT APPROACHES (3)

1) Direct Logit Attribution (DLA)

2) Supervised Probes

3) *Sparse Autoencoders (SAEs)*

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b) Once trained, these **serve as a proxy layer** for analysis of LLM vector space

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1) identify representative ***tokens using logit attribution***

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- followed by *manual* or *automatic* feature naming.



1 DLA limits feature abstraction to *model's tokens*

2 Extracted features are *verbose, broad* and *data-dependent*

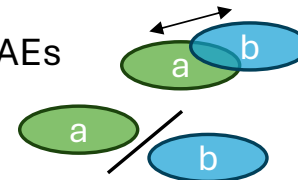
PROBLEMATIC STEP

APPROCH: *Hyperdimensional probe*



Our approach is a **hybrid supervised probe** based on VSAs, taking advantage of the **orthogonality property of high-dimensional vectors** to combine:

- The ability to uncouple superposed subspaces of SAEs
- The interpretability offered by conventional probes



1

COMPARISON with DLA

- No constraints on model's output vocabulary
- It supports sets of concepts with **unrestricted levels of abstraction, cardinality, and data types**.

2

COMPARISON with SAEs

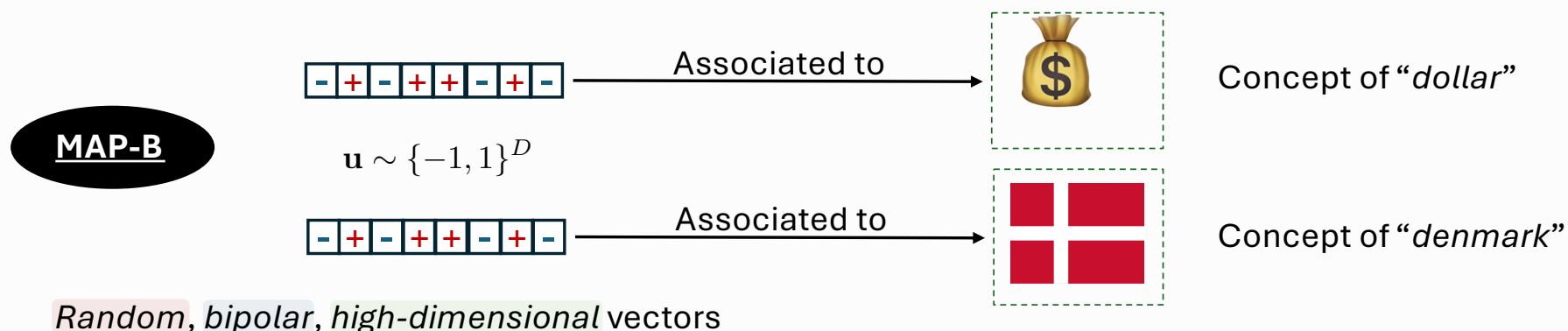
- Our controlled vector space mimics the proxy layer of SAEs **without requiring a subsequent feature-naming step**.
- Analyse **multiple layers simultaneously**, unlike traditional SAEs which target one layer at a time.

Vector Symbolic Architectures

VSAAs are a *computational framework* inspired by cognitive science:

- *neural representations* $\mapsto$ *human-readable symbols*

ASSUMPTION: atomic information can be represented as *random points in a high-dimensional vector space*.*



*Very high-dimensional random vectors have the property of being orthogonal / independent.

Algebra of VSAs

VSA enables **encoding complex concepts** with an apparently **simple algebra**:

1) Build up the alphabet (called codebook)

- Our set of atomic concepts, which can be any possible type of information
- It may consist of domain-relevant concepts such as:
 - $\{PANDA, DOG \dots, CAT\}$ for animal-related inputs.

2) Construct complex concepts using VSA operations:

- a) Binding operator $\odot$
 - Is used to connect two concepts, encoding one concept with a context
 - e.g., key-value $\rightarrow PANDA \odot NOUN$
 - Item-wise *product* (*Hadamard product*)
- b) Bundling operator $+$ (superposition)
 - is used to stack two concepts to create sets (e.g., $CAT + DOG$)
 - Item-wise sum
- c) Unbinding operator $\oslash$
 - Reverse the binding operation, factoring out one vector from another
 - Vector's inverse product

Algebra of VSAs

VSA enables **encoding complex concepts** with an apparently **simple algebra**:

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a) Binding operator $\odot$

- **connects two concepts**, encoding one concept with a context
e.g., key-value pair such as $DOG \odot NOUN$
- Item-wise *product* (*Hadamard product*)

b) Bundling operator $+$, superposition)

- **stacks two concepts** to create sets (e.g., $PANDA + DOG$)
- Item-wise sum

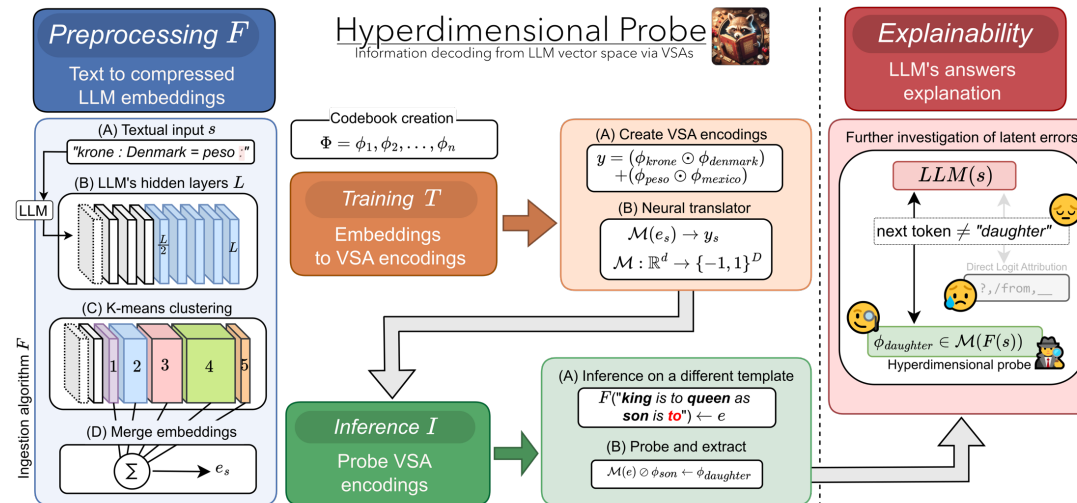
c) Unbinding operator $\oslash$

- **Reverse the binding operation**, factoring out one vector from another
- Vector's inverse product for MAP-B



APPROCH: a three-stage pipeline

OBJECTIVE: Decoding information from latent representations of LLMs using VSAs



1 COMPRESS

The internal state of the language model

2 TRAIN

a supervised neural network to map neural embeddings into a VSA-based proxy space

3 PROBE

this proxy space by extracting concepts from VSA encodings

Explainability

Insights into the semantics encoded in LLMs

Synthetic data

ANALOGIES: textual inputs representing **pairs of concepts connected by the same type of *factual*, *syntactic*, or *semantic relationship*.**

verb+ment_irreg

accomplish : accomplishment = agree : agreement

plural

man : men = road : roads

TARGET

verb_Ving_3pSg

involves : involved = tells : told

currency

USA : dollar = Denmark : krone

male_female

boy : girl = brother : sister

math_div2

44 : 22 = 42 : 21

Table 12: Overview of our experimental set, grouped by tasking an LLM to cluster the domains.

Category		Domains	Docs
Morphological Modifiers	14	noun+less, adj+ness, ...	34,308 (30%)
Verbal & Grammatical Forms	13	past tense, plural, ...	31,219 (27%)
Factual Knowledge	7	country capital, occupation, ...	18,800 (17%)
Semantic Relations	8	family, genders, ...	16,831 (15%)
Mathematics	7	math double, math division5, ...	6,816 (6%)
Semantic Hierarchies	3	hypernyms, hyponyms, ...	6,125 (5%)
	52		114,099 (100%)

Knowledge bases: [Google Analogy Test Set](#) & [The Bigger Analogy Test Set](#)

$a_1 : a_2 = b_1 : b_2$

e.g., LLM("Denmark : krone = Mexico : ")

TO PROBE

TRAINING
STAGE

VSA encodings

PROXY REPRESENTATIONS

For each training input

└─ we generate its **VSA encoding**

└─ by using the **hypervectors of its constitutive words**

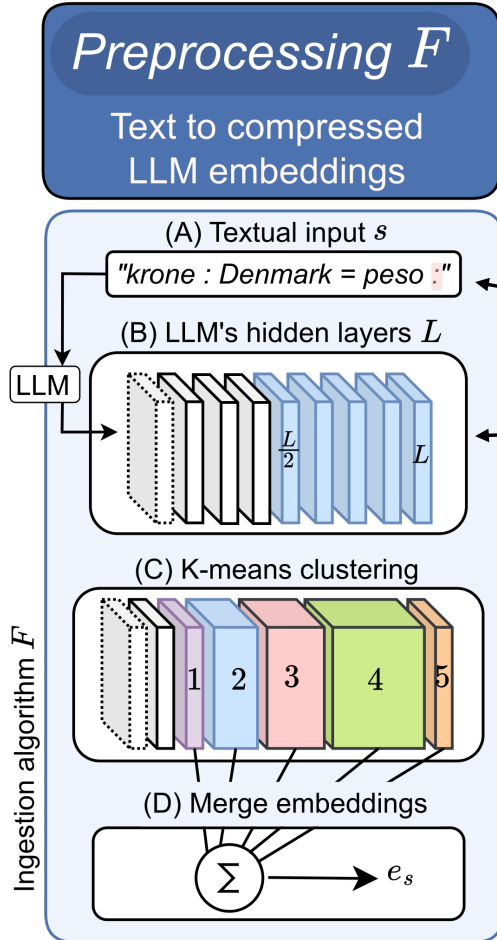
$$y_s = (\phi_{a_1} \odot \phi_{a_2}) + (\phi_{b_1} \odot \phi_{b_2})$$

“_Denmark_: _krone_ = _Mexico_: _peso_” $\mapsto$ $(\phi_{denmark} \odot \phi_{krone}) + (\phi_{mexico} \odot \phi_{peso})$

Binding connects the
key and value of a pair

Bundling creates a stacked set of pairs

(1) Process LLM embeddings



Algorithm 1: Ingestion procedure F

Data: Textual sequence $s \in \mathcal{S}$

Result: Compressed model state for its next token prediction

begin

// Get the residual stream from the language model
 $\mathbf{H} \leftarrow \text{LLM}(s) \in \mathbb{R}^{L \times T \times d}$;

// Retain embeddings of the last token from the bottom half of the layers
 $\mathbf{H}^* \leftarrow \mathbf{H}[L/2 : L, -1]$;

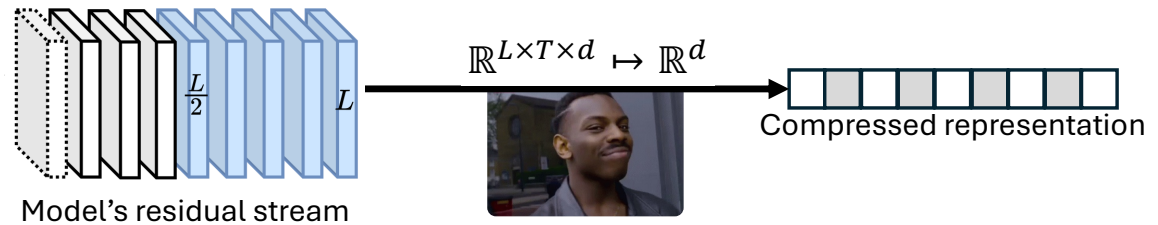
// Apply K-Means clustering

$\mathbf{C} \leftarrow \text{KMeans}_{K=5}(\mathbf{H}^*) \in \mathbb{R}^{K \times d}$;

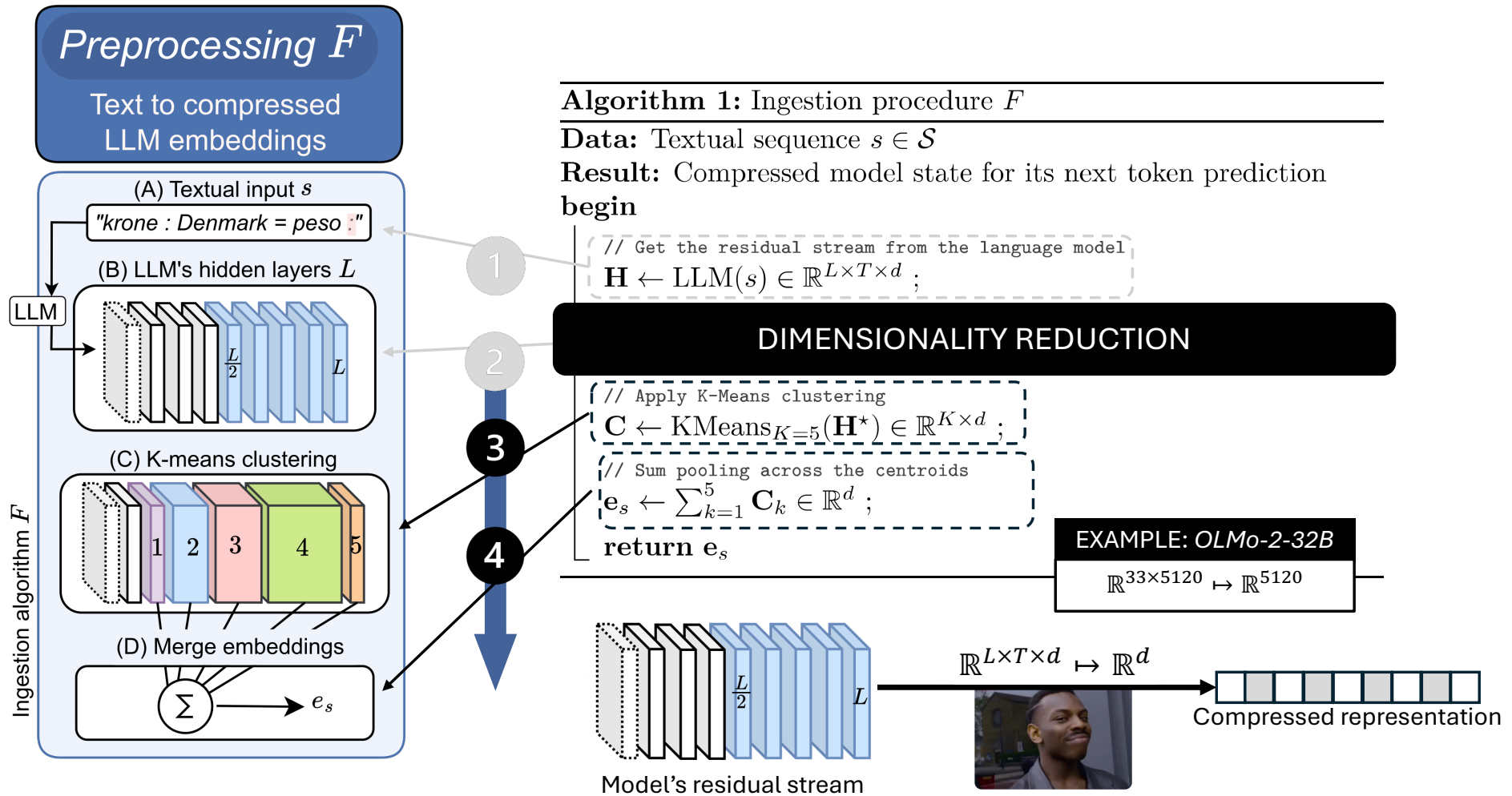
// Sum pooling across the centroids

$\mathbf{e}_s \leftarrow \sum_{k=1}^5 \mathbf{C}_k \in \mathbb{R}^d$;

return $\mathbf{e}_s$



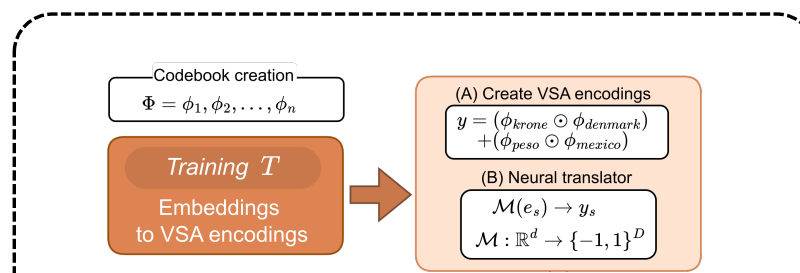
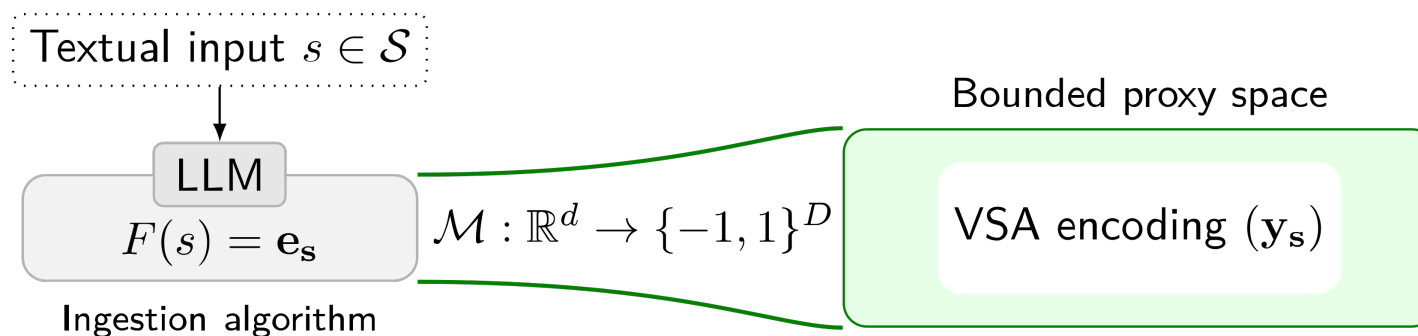
(1) Process LLM embeddings





(2) Train a neural VSA encoder

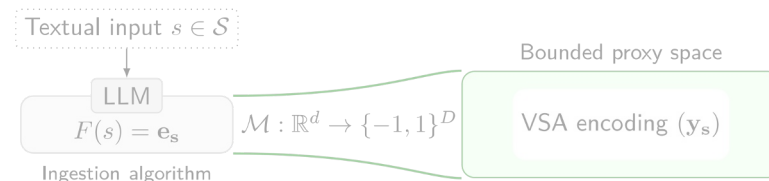
We define the *neural VSA encoder model* as a **three-layer MLP with 55M–71M parameters**, performing the non-linear transformation:





(2) Train a neural VSA encoder

We define the *neural VSA encoder model* as a **three-layer MLP with 55M–71M parameters**, performing the non-linear transformation:



TRAINING PERFORMANCE

Average accuracy of 93 %

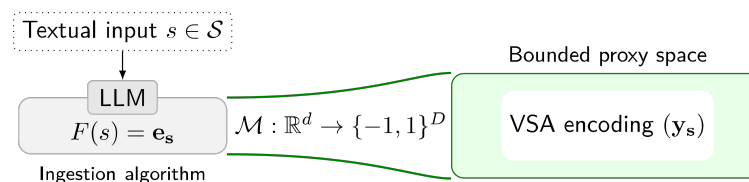
a) Strong numerical alignment between our encoder's outputs and the target encodings

b) Robust classification accuracy after polarizing the predictions with the sign function

Table 4: Training performance of our neural VSA encoder $\mathcal{M}$ on the test set. Order by model size.

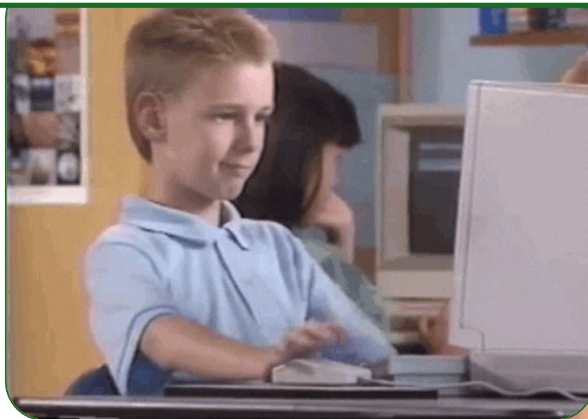
Large Language Model				Cosine similarity	Binary accuracy
Name	Parameters (B)	Embedding dimension	Layers from residual stream		
Llama 4 - Scout, 17B-16E	109	5120	24 th to 48 th 25	0.890	0.934
OLMo-2	32	5120	32 nd to 64 th 32	0.878	0.926
Phi 4	14	5120	20 th to 40 th 21	0.881	0.930
Llama 3.1-8B	8	4096	16 th to 32 nd 17	0.892	0.937
Pythia-1.4b	1.4	2048	12 th to 24 th 13	0.861	0.916
GPT-2, medium	0.4	1024	12 th to 24 th 13	0.865	0.920
AVERAGE				0.878 ± 0.01	0.927 ± 0.01

(2) A neural VSA encoder

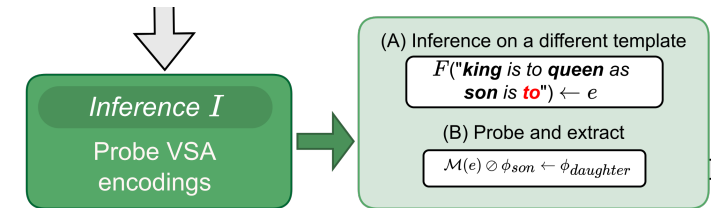


PROXY
LAYER

LLM embeddings can be accurately represented
using fully distributed **encodings in VSAs**



(3) Probe VSA encodings



TASK

We **examine the VSA encodings** produced by our trained neural VSA encoder

OBJECTIVE

Query the generated encodings to **understand which concepts are encoded**, probing indirectly the LLM embeddings

HOW

We **retrieve the atomic concepts** embedded in a VSA encoding using the unbinding operation

Reverse the binding

For example, **unbinding a VSA encoding with the concept of Mexico and obtaining Peso** suggests that the probed encoding originally incorporated both the key and value of the target analogy pair:

LET $s := \text{"Denmark:krone=Mexico:"} \mapsto \text{"peso"}$
 COMPUTE $y_s = \mathcal{M}(F(s))$
 QUERY $y_s \oslash \phi_{\text{mexico}} = \phi_{\text{peso}} + \text{noise}$
 THEN $y_s \approx (\phi_{\text{mexico}} \odot \phi_{\text{peso}})$



Experiments

We validate our methodology using textual *inputs with syntactic structures that differ from those seen during our training stage*:

$a_1 : a_2 = b_1 : b_2$

e.g., Denmark : krone = Mexico : ↦ peso

TRAINING
STAGE



TO PROBE

a_1 *is to* a_2 *as* b_1 *is to* b_2

e.g., king is to queen as son is **to** ↦ daughter

EXPERIMENTAL
STAGE

MOTIVATION

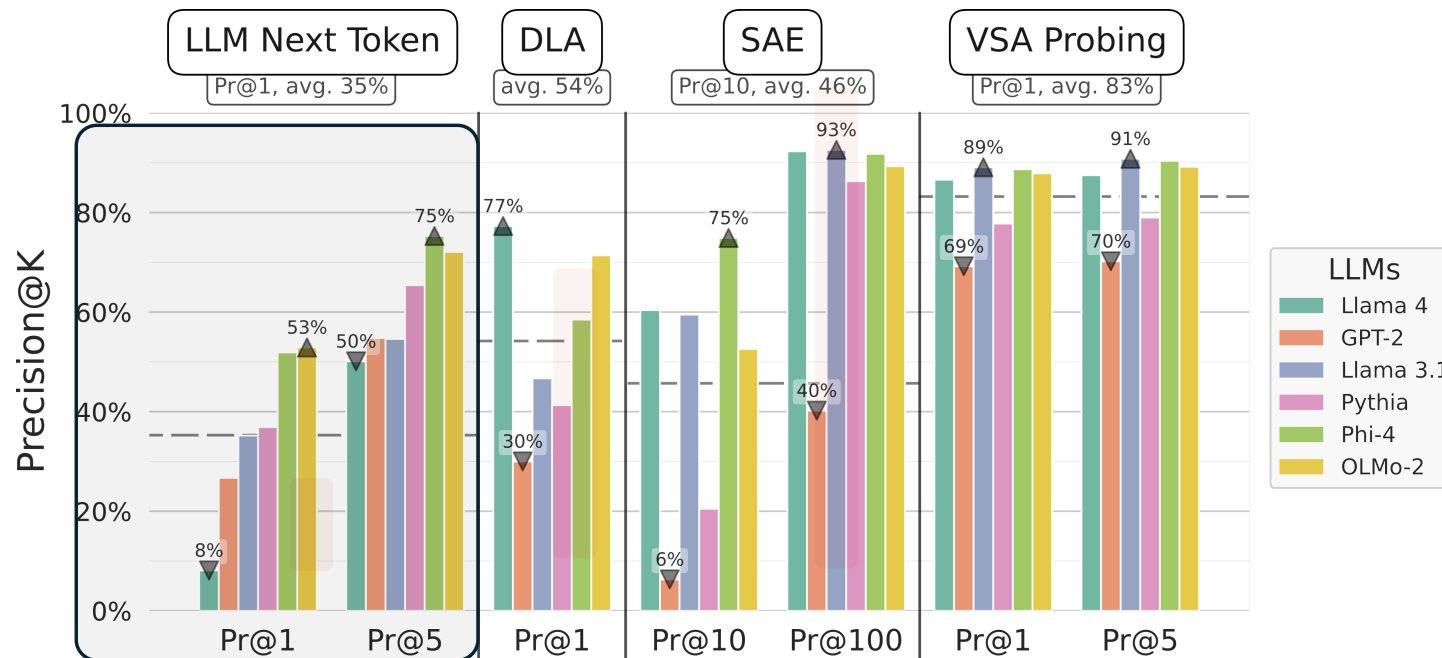
To further mitigate confounding effects from probe learning in relation to information decoding.

We want to retrieve **encoded concepts** rather than **memorizing vector transformations**

FINDINGS: *LLM Performance*

High variability in LLM performance: the largest model (109B; *Llama 4, Scout*) exhibited the lowest precision@1 in the next-token prediction task

- 1) even underperforming the legacy GPT-2
- 2) Its next-token@5 was comparable to others (still the lowest),
- 3) Ranked among the best in probing@1.



The **final state encodes the target**, but the model fails to predict it as next token.

Likely influenced by endogenous and exogenous factors

This emphasizes the variability introduced by *tokenization* and *prompt design* in token-oriented analysis

FINDINGS: VSA Probing

VSA exposes *varying conceptual richness*

Table 1: **Concepts extracted though hyperdimensional probe.** Key|Target indicates extraction of both the key (b_1) and value (b_2) of the target analogy; Key for only b_1 . Example refers to a_1 and a_2 , the concepts of the in-context example; Context|Target for all four concepts. Key Values indicates concepts linked in a different domain, Out-of-context for those unrelated to input. NONE means no concepts. *On average, our probe can capture target concepts in 78% of all the analogies.*

Extracted Concepts (%)	GPT-2	Pythia	Llama 3.1	Phi-4	OLMo-2	Llama4, Scout	AVERAGE
Key Target	60.0	66.9	85.4	84.8	80.1	79.0	76.0 ± 9.4
NONE	21.0	16.7	6.0	7.6	8.5	11.5	12.2 ± 5.4
Key	EXAMPLE			1.0	1.8	3.4	2.9 ± 2.0
Example	KEY	2.4	VALUE	1.3	KEY	0.8	TARGET
Context Target	0.0	0.0	1.2	1.2	2.4	2.4	
E.g., {son, daughter} Key Values	“king is to queen as son is to” $\mapsto$ “daughter”						
Out-of-context							
Example Value Key Values	1.5	1.3	0.3	0.0	0.2	0.1	0.6 ± 0.6
Key Values Target	0.4	0.1	0.3	0.4	0.1	0.1	0.2 ± 0.1
Target	0.1	0.1	0.1	0.2	0.1	0.1	0.1 ± 0.0

FINDINGS: VSA Probing

probing exposes *varying conceptual richness*

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NONE	21.9	16.7	6.9	7.6	8.5	11.5	12.2 ± 5.4
Key	4.5	6.1	0.6	1.0	1.8	3.4	2.9 ± 2.0
Example	5.8	2.4	1.5	1.3	2.0	0.8	2.3 ± 1.6
Context Target	1.1	1.9	1.5	1.2	4.4	2.4	2.1 ± 1.1
Key Key Values	1.3	1.4	1.8	1.5	1.1	0.8	1.3 ± 0.3
Out-of-context	1.6	1.2	0.5	0.7			
Example Value Key Values	1.5	1.3	0.3	0.0			
Key Values Target	0.4	0.1	0.3	0.4			
Target	0.1	0.1	0.1	0.2			

“king is to queen as son is to” $\mapsto$ “daughter”
 GPT-2 $\rightarrow \{\phi_{king}, \phi_{queen}\}$
 OLMo-2 $\rightarrow \{\phi_{king}, \phi_{queen}, \phi_{son}, \phi_{daughter}\}$

Struggle with the NLP task of completing analogies, even when they **appear to grasp the in-context concepts**

It exhibits **richer representations in its final state** for the concepts of both the *input context* and *next word*

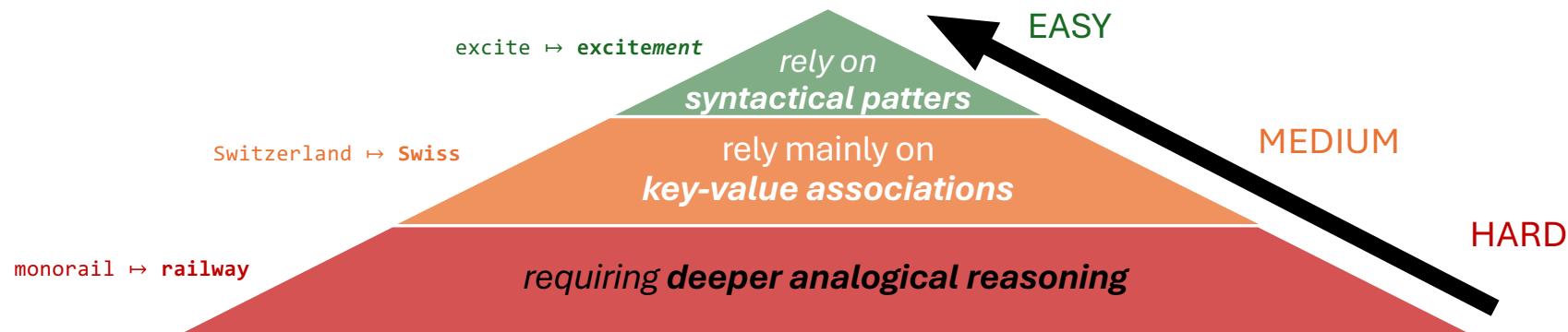
Also, the best in @next token@1

FINDINGS: VSA Probing

probing exposes *varying conceptual blankness*

Table 7: Analogies by Area (%) for the subset of instances with no retrieved concepts for Llama 4 and OLMo2, mentioned in Section 5.2. OLMo-2 shows richer embeddings than Llama 4, with lower proportions of instances with conceptually-blank representations for most of the areas. Llama 4 slightly outperform OLMo2 in mathematical and grammatical analogies.

Area	Llama 4 (docs, %)	OLMo2 (docs, %)	Sample	Domain
Mathematics	87.8	91.1	80 is to 160 as 98 is to	math double
Semantic Hierarchies	38.8	30.8	limousine is to car as monorail is to	hyponyms
Semantic Relations	10.0	3.9	Croatia is to Croatian as Switzerland is to	nationality adjective
Factual Knowledge	5.5	5.1	euclid is to Greek as galilei is to	name nationality
Verbal & Grammatical Forms	1.4	2.1	seeing is to saw as describing is to	past tense
Morphological Modifiers	1.1	0.8	agree is to agreement as excite is to	verb+ment



FINDINGS: VSA Probing

Concept drift during text generation

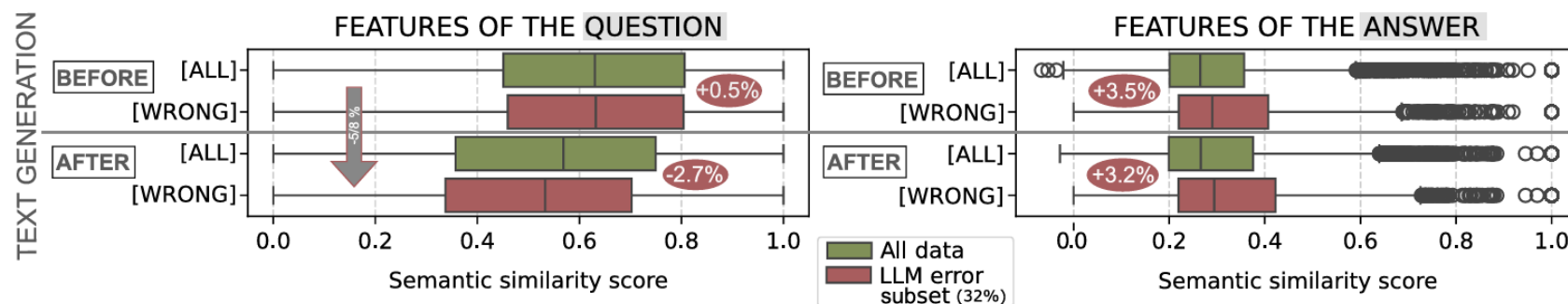


Figure 5: Concepts extracted *before* and *after* the LLM's text generation, with respect to *question* and *answer* features. Red denotes the subset of failure instances, while green the full sample $\bar{Q}$.

Focus on LLM failure instances → analysis of the *extracted concepts related to questions and answers*

- **Question-related concepts:** slight decrease after text generation (-2.7%)
- **Answer-related concepts:** slight increase before and after text generation (3.2 – 3.5 %)

This suggests that LLM failures may stem from losing focus on the question rather than from a lack of answer-related knowledge

CONCLUSIONS



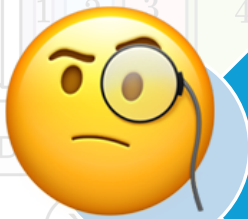
VSA for neural probing: latent features of LLM vector space can be expressed through VSA encodings and hypervector algebra



Hyperdimensional probe: a novel probing paradigm that performs input-aligned feature extraction for output-oriented interpretability



Effective compression of LLM embeddings to **probe the model's complete internal state**, removing the need for the layer-selection stage common in earlier work



Enhanced interpretability of neural representations: concept-oriented insights into LLM inference of diverse input types and during text generation





THANKS

Open to collaboration on **VSA**s applied to **LLMs**



LinkedIn

Bronzini, M., Nicolini, C., Lepri, B., Staiano, J., & Passerini, A. (2025).
*Hyperdimensional Probe: Decoding LLM Representations via
Vector Symbolic Architectures*